

Paper Manager pack — V-JEPA 2.1 (Nov 30 · Accompanying Models)

Paper: V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning
arXiv: <https://arxiv.org/abs/2603.14482> · [code](#) · arXiv **2603.14482v3** (11–12 Jun 2026)
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PDF: </workspace/papers/vjepa-2.1-2603.14482.pdf> (= </workspace/vjepa21-quiet/paper.pdf>) · **37 pp** · ~37.4 MB
Session: Nov 30 · due 9:30 AM · Type: **Accompanying Models (World Models, Reasoning, Planning)**
Role weights: **Archaeologist=2 (Lukas Moroz) LONGEST PRIMARY**; Stakeholder / Reviewer / Empiricist / Visionary seats empty — **still all five produced**; Archaeologist heaviest
Kanta role: **NOT seated** this session — still synthesize Archaeologist-richest five-role pack for Lukas; deepen CLEAR JEPA lineage through **2.1**; world-model / dense / planning deep dive; Empiricist course-smoke run pack
Themes: dense predictive loss `L_ctx` · deep multi-level self-supervision · multi-modal tokenizers · VisionMix163M · keep global motion (SSv2) · Ego4D STA / EK anticipation · Franka AC grasp · latent nav vs NWM · lineage CLEAR I-JEPA → V-JEPA → V-JEPA 2 / 2-AC → **2.1** → 2-AC on 2.1 encoder
Sources: Summarizer (</workspace/papers/vjepa-2.1-2603.14482-summary.md>) + Senior Research (</workspace/vjepa21-quiet/DELIVERABLE.md>) + Ablation Bot (</workspace/papers/vjepa-2.1-2603.14482-ablations.md>) — **AUTHORITATIVE** + [paper_notes.md](#) / [env_outline/](#)
Sibling links: </workspace/ijepa-quiet/> (heavy lineage) · </workspace/siglip-quiet/> · </workspace/pi05-quiet/> · </workspace/openpi-comet-smoke/> · </workspace/diffusion-policy-quiet/> · </workspace/droid-quiet/> · </workspace/data-scaling-laws-quiet/>

1. One-liner

Unlock dense video features without abandoning global motion — add distance-weighted **context loss `L_ctx`** (supervise visible tokens) + **deep multi-level self-supervision**; then scale (VisionMix163M, ViT-G, cool-down). Fixes V-JEPA 2's weak dense maps (ADE **22.2** → **47.9**, NYU **0.682** → **0.307**) while keeping SSv2 **77.7**; transfers to Ego4D STA **7.71**, Franka AC grasp **60** → **70** → **80%** (n=10), and latent nav ~**10×** faster vs NWM (**10.6 s** vs **103.2 s**).
Contrast poles (consistency triangle):

Pole	What they do	Primary claim / use
DINOv2 / DINOv3 / SigLIP	Discriminative JEA / dense image features	VLA vision towers (π_0 , OpenVLA, ...); dense image SOTA foil
MAE / VideoMAE	Generative MIM — reconstruct pixels	Strong FT; weaker frozen linear
I-JEPA → V-JEPA → 2 → 2.1 → 2-AC	Predictive embedding / latent world model	Semantic SSL → video → dense+temporal → action-conditioned planning

Do NOT invent / claim: BEHAVIOR hybrid JEPA+FM as an **in-paper** result (proposal only); Franka as general manipulation SOTA (cup table-top, n=10/skill); ADE “beats DINOv3” (NYU does; ADE **47.9** trails DINOv3-7B **55.9**); course smoke = ViT-G / Franka 80% / NYU 0.307; π_0 /OpenVLA ship V-JEPA towers.

2. Summary (exec overview + key numbers)

Mur-Labadia et al. (arXiv 2603.14482 / Meta FAIR) diagnose why V-JEPA 2 is strong on global motion but weak on dense spatial structure: classic `L_predict` supervises **masked tokens only**, so context tokens behave like **register / global aggregators** → noisy PCA, ADE **22.2**, NYU **0.682**. Fix = four ingredients: (i) **Dense Predictive Loss `L_dense`**

= $L_{\text{predict}} + L_{\text{ctx}}$ (distance-weighted L1 on nearby context); (ii) **Deep Self-Supervision** (multi-level predictor over 4 encoder levels); (iii) **Multi-Modal Tokenizers** (native 2D image / 3D video + modality tokens); (iv) **scale** — VisionMix163M, ViT-G 2B, high-res cool-down + distill to ViT-L/B.

Why Accompanying Models session: Unlike I-JEPA (image-only, zero robots), **this PDF is both** an SSL methods paper **and** a world-model robotics paper — AC Franka CEM + latent nav are **in-paper**. Continuity from Nov 25 I-JEPA: students see AMI → JEPA → video → **dense state** → planning.

Axis	Number / claim
Venue / code	arXiv 2603.14482v3 ; facebookresearch/vjepa2 CLEAR
Recipe stack ADE (T1)	VJ2 22.2 → + L_{ctx} 33.8 → +deep SS 38.6 → full 47.9
Recipe stack NYU↓ (T1)	0.682 → 0.474 → 0.463 → ... → 0.307
Recipe stack SSv2 (T1)	72.8 → + L_{ctx} 62.5 (hurt) → +deep SS 72.1 → full 77.7
L_{ctx} weighting (T2)	weighted+warmup ADE 33.8 / SSv2 62.5 (chosen); λ_{final} 0.5 vid / 0.7 img
Ego4D STA mAP All (T4)	ViT-G 7.71 (~+35% vs STAformer 5.67 ; VJ2-g 6.02)
EK100 Action R@5 (T5)	ViT-G 40.8 (VJ2-g 39.7 ; 2.1-g 38.4 — needs G scale)
Franka Grasp (T6, n=10)	VJ2 60% → 2.1 matched 70% → 2.1 H=8 80%
Nav Tartan ATE / time (T7)	ViT-G 5.687 ATE; 10.6 s vs NWM 103.2 s (~10×)
Dense vs DINOv3 (T8)	NYU 0.307 beats DINOv3-7B 0.309 ; ADE 47.9 trails 55.9
Global SSv2 (T9)	ViT-G 77.7 SOTA frozen attentive; 2.1-g 76.9 slightly under VJ2-g 77.3
Distill ViT-L (T11)	SSv2 76.5 / ADE 46.7 / NYU 0.336 ≈ teacher
Primary train	135k steps; 16f@256; batch 128 clips + 2304 images; EMA 0.99925
Cool-down	12k; video 64f@384; image 512; NYU 0.365 → 0.307

Must-cite tattoo: NYU **0.307** · ADE **22.2** → **33.8** → **38.6** → **47.9** · Ego4D STA **7.71** · grasp **60 / 70 / 80** (n=10) · nav ATE **5.687** + **10.6 s** vs **103.2 s** · L_{ctx} vs deep SS trade-off (SSv2 **72.8** → **62.5** → **72.1**).

Do NOT claim: BEHAVIOR hybrid as demonstrated; course smoke = paper absolutes; ADE beats DINOv3; +20% grasp without clarifying matched (+10 pp) vs longer-horizon (+20 pp); temporal-mask as paper ablation (fixed [1.0,1.0]).

3. Keywords

V-JEPA 2.1; dense predictive loss; context loss L_{ctx} ; deep self-supervision; multi-modal tokenizer; VisionMix163M; video SSL; joint-embedding predictive architecture; world models; action-conditioned prediction; CEM; Franka; Ego4D STA; EPIC-KITCHENS anticipation; NYUv2 depth; ADE20K; SSv2; Navigation World Model; I-JEPA lineage; Accompanying Models; BEHAVIOR backbone ablation

4. Deep dive — World-model / dense / planning (PRIMARY SESSION CONTRIBUTION)

Callout: Scientific product of *this* PDF = **dense+temporal latent prediction** that upgrades the same AC planner / nav stack used in V-JEPA 2. Session type = Accompanying Models because agents that predict & act over continuous time need **dense state estimation**, not only global video embeddings.

What is “world-model-ish” here — CLEAR (in-paper)

Piece	In-paper role	World-model reading
Predictor \ (P_{\phi})	24-block ViT emb 384; context + mask tokens → multi-level emb preds	Conditional latent mask-denoiser \ (P(\text{latent}_{target} \mid \text{latent}_{context}, z_{\text{pos}})) — spatio-temporal
L_ctx + deep SS	Supervise all tokens at 4 encoder levels	Dense state that preserves local geometry / depth for MPC
AC predictor (§3.3)	Freeze 2.1 enc; 24-layer ~300M action-cond. TF on DROID ; CEM image-goal MPC	Same 2-AC recipe as Assran’25 with encoder swap to 2.1
Grasp T6	60% → 70% matched CEM → 80% H=8	Dense/depth unlocks longer rollouts (paper: VJ2 degrades with longer horizon)
Nav CDiT (§3.4)	Clean-pred + DDIM on 2.1 reps; CEM N=480	Latent nav WM; ~ 10× vs SD-VAE NWM (10.6 s vs 103.2 s); Tartan ATE 5.687
Anticipation T4/T5	Ego4D STA 7.71 ; EK Action 40.8	Predictive reps for embodied forecast

What is not overclaimable

- Franka = cup Reach/Grasp/Pick-&-Place; **n=10** tasks/skill avg; lab table-top — not BEHAVIOR-1K / OXE.
- Nav = open-loop **2 s @ 4 FPS** on Tartan/Scand/Sacson — not full autonomy.
- Nav **10×** changes representation **and** sampler (8 DDIM vs ≥128) — joint system claim.
- Remaining Grasp/P&P failures = **gripper timing**, not spatial misunderstanding (paper attribution).
- Hybrid JEPA imagination + FM motor for BEHAVIOR = **Visionary proposal**, not an in-paper result.
- Temporal mask **not** ablated (fixed tube [1.0,1.0] = V-JEPA 2).

CLEAR lineage path (cite these labels; do not rewrite I-JEPA pack)

Work	What it adds	Label
I-JEPA (2301.08243); ijepa-quiet/	Image JEPA; multi-block; emb L2	CLEAR parent
V-JEPA (2404.08471)	Video feature prediction	CLEAR
V-JEPA 2 (2506.09985)	Internet-scale + first 2-AC Franka CEM (≤62 h DROID)	CLEAR
V-JEPA 2.1 (this, 2603.14482)	Dense unlock + better AC/nav with 2.1 encoder	CLEAR (was TENTATIVE in I-JEPA pack)
V-JEPA 2-AC on 2.1	§3.3 reuses Assran’25 AC codebase; encoder swap	CLEAR extension
facebookresearch/vjepa2	ViT-g/G + distilled B/L	CLEAR

Design rhyme vs VLA towers (TENTATIVE — do not overclaim)

π_0 / $\pi_{0.5}$ / OpenVLA / GR00T packs **do not cite** V-JEPA 2.1. Their towers = SigLIP/DINOv2; “predictive” = FM/diffusion action experts. Architecture **rhyme** only. BEHAVIOR’25 winners ride $\pi_{0.5}$ + **FM** + **SigLIP** — **parallel** world-interface bet, not JEPA child.

World-model exam bite: "V-JEPA 2.1 = dense-state upgrade to video JEPA: L_ctx +deep SS unlock NYU **0.307** / ADE **47.9** while SSv2 stays **77.7**; same AC CEM planner → grasp **60** → **70** → **80** (n=10); nav **10.6 s** vs **103.2 s**. Robotics inventum = **state estimation**, not a new planner — cite T6/T7 carefully; do **not** invent BEHAVIOR hybrid as in-paper."

5. Deep dive — Dense / SSL recipe (L_ctx + deep SS)

Callout: Fig. 3 + Tables 1–2 are the session’s methods spine. Mask-only JEPA → context-as-register → weak dense; supervise visible tokens (weighted) + hierarchical loss restores Pareto.

Architecture checklist (Empiricist pin)

Piece	Spec (paper default)
Image tokenizer	2D conv 16×16
Video tokenizer	3D conv 16×16×2 (tubelet=2) + modality emb + 3D RoPE
x-encoder $\backslash(E_{\backslash\theta})$	ViT-B/L/g/G; multi-level outs → MLP fuse
y-encoder $\backslash(E_{\backslash\bar{\theta}})$	EMA mom 0.99925 ; full unmasked; multi-level targets
Predictor $\backslash(P_{\phi})$	24 blocks, emb 384 ; multi-level preds
Losses	L1; $L_{\text{dense}} = L_{\text{predict}} + L_{\text{ctx}} ; \backslash(\lambda_i = \lambda / \sqrt{d_{\min}(i,M)})$
Deep SS levels	4 (ViT-G indices [12,24,36,48])
Mask	Spatial [0.15,0.7]; aspect [0.75,1.5]; temporal [1.0,1.0] (= VJ2; not ablated)
Primary / cool-down	135k @ 16f/256; +12k @ 64f/384 + img 512
Final λ	0.5 video / 0.7 image (T12)

Ablation Bot **AUTHORITATIVE** — must-hit deltas

Ablation	Paper locus	Headline numbers
+ Context Loss	T1 · Fig 3	ADE 22.2 → 33.8 ; NYU 0.682 → 0.474 ; SSv2 72.8 → 62.5 ; IN 82.2 → 72.6
λ weighting	T2	weighted+warmup 33.8 / 62.5 best; fixed λ=0.5 → ADE 27.5 / SSv2 53.8
+ Deep SS	T1 · T13	SSv2 62.5 → 72.1 ; ADE 33.8 → 38.6 ; NYU 0.463 ; Last vs 4-layer gap shrinks
+ VisionMix	T1 · T3	ADE 38.6 → 40.8 ; NYU 0.418
+ Multi-modal Tok	T1	ADE 40.8 → 41.4 ; recognition flat
+ Scale + cool-down	T1 · T14	ADE 47.9 ; NYU 0.307 ; SSv2 77.7
Robot grasp	T6	60 / 70 / 80 (n=10); H=8 helps 2.1
Nav vs NWM	T7	ATE 5.687 ; 10.6 s vs 103.2 s
Temporal mask	—	No paper ablation — extension only

Taxonomy / recipe exam bite: "L_ctx unlocks dense (ADE **22.2 → 33.8**, NYU **0.682 → 0.474**) but nukes globals (SSv2 **→10**); deep SS restores SSv2 **72.1** and lifts ADE **38.6**; full recipe NYU **0.307** / SSv2 **77.7** / ADE **47.9**. Temporal mask fixed — do not claim as ablation."

6. Role-ready sections (all five — Archaeologist RICHEST)

6.1 Archaeologist — **WEIGHT 2 LONGEST PRIMARY** (Lukas Moroz)

Seat charge: Deepen **CLEAR / TENTATIVE** lineage I-JEPA → V-JEPA → V-JEPA 2 / 2-AC → **V-JEPA 2.1** → 2-AC on 2.1 encoder. Map consistency triangle vs DINOv3/SigLIP and MAE/VideoMAE. Fence robot claims to **Tables 6–7** (n=10; system confounds). Longest talk block (~16 slides). **Kanta not seated** — Lukas carries lineage.

Priors — CLEAR (cited / direct parents)

Prior	Role relative to V-JEPA 2.1	Label
LeCun 2022 AMI	Names JEPA; predictive embeddings	CLEAR
I-JEPA Assran'23 (2301.08243); ijepa-quiet/	Image JEPA; multi-block; emb loss; EMA	CLEAR
V-JEPA Bardes'24 (2404.08471)	Video feature-prediction JEPA	CLEAR
V-JEPA 2 Assran'25 (2506.09985)	Scale + 2-AC planning recipe; immediate parent	CLEAR
BYOL / stop-grad + EMA	Collapse avoidance reused	CLEAR
DINO / DINOv2 / DINOv3	Dense-feature foil & table baselines	CLEAR (cited)
Registers Darcet'23	Explains context-as-global-aggregator hypothesis	CLEAR
LVD-142M / DINOv2 curation	Image half of VisionMix163M	CLEAR
NWM Bar'25	Nav baseline (SD-VAE); this paper swaps latent	CLEAR
DROID Khazatsky'24	AC post-train robot data	CLEAR
data2vec / MAE / VideoMAE	Mask-predict / generative foils	CLEAR

This paper — CLEAR inventum

- 1. **Diagnosis:** mask-only JEPA → weak dense maps (ADE 22.2 / NYU 0.682).
- 2. **Dense Predictive Loss:** supervise all tokens; distance-weighted L_ctx near masks.
- 3. **Deep Self-Supervision:** multi-level predictor restores globals while lifting dense.
- 4. **Native multi-modal tokenizers** (fix image-as-static-video hack).
- 5. **VisionMix163M** rebalance (LVD-142M in; ImageNet out; YT/SSv2 up).
- 6. **Empirical package:** STA 7.71, EK 40.8, NYU 0.307 (beats DINOv3-7B), SSv2 77.7; grasp 60 → 80; nav ~10x.
- 7. Distilled ViT-L/B family for deployability.

Subsequent — CLEAR

Work	What it adds	Label
facebookresearch/vjepa2 2.1 weights	Reproducible encoders + distillates	CLEAR
V-JEPA 2-AC recipe on 2.1 encoder	This paper §3.3	CLEAR extension

Subsequent / parallel — TENTATIVE (do not overclaim)

Work	Relation	Label
π_0 / $\pi_{0.5}$ / OpenVLA / GR00T	No V-JEPA cites; SigLIP/DINO towers	Architecture rhyme only
BEHAVIOR winners ($\pi_{0.5}$ stacks)	FM + SigLIP — parallel WM bet	TENTATIVE as “same agenda”
Hybrid JEPA-rollout + FM expert	Research proposal for BEHAVIOR	TENTATIVE — not in-paper
DINO-Foresight / dense → WM adjacent	Cited adjacent research	TENTATIVE sibling agenda
Point-cloud / other-modality JEPA	Pending local PDFs	TENTATIVE

Genealogy diagram (CLEAR framing)

Predictive coding / LeCun AMI / JEPA abstract

→ I-JEPA (Assran 2023) – image multi-block latent prediction

[pack: /workspace/ijepa-quiet/]

→ V-JEPA (Bardes 2024) – video mask-denoising JEPA

→ V-JEPA 2 (Assran 2025) – scale + AC world-model / robot CEM

→ **V-JEPA 2.1 (this paper, 2026)** – dense loss + deep SS + multimodal tok

→ frozen backbone for anticipation / depth / seg / VOS

→ AC predictor (DROID) + CEM manipulation; CDiT nav WM

→ optional 2026 BEHAVIOR video / WM backbone *candidate*

(Empiricist probes; not default ship)

Consistency triangle (session tattoo)

DINOv2/v3 / SigLIP — discriminative dense (image-first) — VLA towers
MAE / VideoMAE — generative pixels — FT-heavy
I-JEPA→V-JEPA→2→2.1→2-AC — predictive latents + now dense — planning WM

Deepen-toward-robotics note (Lukas / Archaeologist)

V-JEPA 2 proved **action-conditioned latent planning** can zero-shot Franka with image goals. V-JEPA 2.1's inventum for robotics is **not a new planner**, but **state estimation quality**: depth/local structure that makes CEM over longer horizons useful (**80%** grasp) and latent nav **~10×** cheaper than VAE latents. For BEHAVIOR / VLA groups: ask whether swapping SigLIP for frozen V-JEPA 2.1 **hurts language grounding** but **helps contact geometry / depth / tracking** — open, measurable. **Do not invent** robot claims beyond Tables 6–7; **do not** present JEPA+FM hybrid as demonstrated.

What to cite V-JEPA 2.1 for (2026 talks)

- Dense + temporal SSL in one JEPA recipe.
- Context-loss insight (supervise visible tokens).
- Deep hierarchical JEPA supervision.
- Frozen-backbone anticipation + depth SOTA-ish numbers (NYU **0.307**).
- World-model planning hooks (grasp / nav) **with** honest N and system confounds.
- **Not** for: drop-in OpenPI SigLIP replacement without probe evidence; full BEHAVIOR win claims; “temporal mask ablation.”

Archaeologist exam bite: “V-JEPA 2.1 (Mur-Labadia’26) = dense unlock for video JEPA: `L_ctx` ADE **22.2** → **33.8** / NYU **0.682** → **0.474** but SSv2 **–10**; deep SS restores **72.1** + ADE **38.6**; full NYU **0.307** / STA **7.71** / grasp **60/70/80** (n=10) / nav **10.6 vs 103.2** s. CLEAR line I-JEPA → V-JEPA → 2 → **2.1** → 2-AC-on-2.1; triangle = DINOv3/SigLIP | MAE | JEPA → 2.1 → 2-AC. **Do not invent** BEHAVIOR hybrid as in-paper.”

6.2 Stakeholder — produced despite empty seat

Buy / why schedule this for “Accompanying Models”:

- Direct continuation of Nov 25 I-JEPA into **temporal + dense + planning** — full AMI → JEPA → robot path in one syllabus arc.
- Rare dual paper: SSL methods **and** world-model robotics (Franka AC + nav) in the **same PDF**.
- Dense unlock answers the V-JEPA 2 objection (“great SSv2, useless depth/seg”) — NYU **0.307** competes with DINOv3-7B.
- Open code (`vjepa2`) + distilled B/L → adoptable.
- Fits theme: agents over continuous time need **dense** state, not only global video embeddings.

Risks / costs / limits:

- Pretrain = Meta-scale (ViT-G 2B, VisionMix163M, 135k+12k, batch 2304 images) — use released weights; not weekend repro.
- Franka: **n=10/skill**, single cup table-top — promising but narrow; don't oversell as general manip SOTA.
- ADE/Cityscapes still trail DINOv3 — cluttered-scene gap admitted.
- At matched 1B, 2.1-g slightly under VJ2 on SSv2 (**76.9 vs 77.3**) and EK Action (**38.4 vs 39.7**) — gains need ViT-G.
- VidQA mixed; TemporalBench/TOMATO regressions on same data.
- VLA market still on SigLIP/DINO — JEPA wins **WM/planning** track, not “drop-in VLM tower.”
- **Kanta not seated** — Archaeologist (Lukas) carries lineage; syllabus owner still decides adoption.

Decision ask: **Required** for world-model / planning track after I-JEPA + V-JEPA 2; optional skim for pure VLA-backbone bake-offs unless evaluating depth/anticipation probes. Pair: V-JEPA 2 (2506.09985) § on 2-AC + this paper

for dense unlock.

Stakeholder exam bite: "Buy V-JEPA 2.1 as the **dense-state upgrade** + in-paper AC/nav demos; buy released ViT-L distill for deploy — don't buy it as a silent SigLIP replacement or as BEHAVIOR SOTA from citing 0.307 alone."

6.3 Scientific Reviewer — produced despite empty seat

Claims under review

1. Supervising **context tokens** (`L_ctx`) unlocks dense spatial structure in video JEPA.
2. **Deep SS** recovers global motion understanding after the context-loss hit.
3. Dense 2.1 features improve **zero-shot AC planning** (grasp) and enable **faster latent nav**.
4. Recipe scales to ViT-G + cool-down with competitive dense+global Pareto.

Strengths

1. Clean causal story: context-as-register hypothesis → `L_ctx` ablation → PCA + linear probes → recipe stack (Fig. 5 / T1).
2. Reports **both** dense and global; admits trade-offs when adding `L_ctx` alone.
3. Robotics reuses public AC codebase + same task configs as V-JEPA 2 — fairer encoder swap.
4. Dense eval protocol matched to DINOv3 — credible NYU claim.
5. Distillation shows 2B is not vanity scale (ViT-L ≈ teacher).
6. Ablation Bot corpus rich (T1/T2/T13 + robot/nav).

Weaknesses / ask-for-revision style

1. Robot $n=10$ /skill; no CIs / multi-object / multi-env.
2. Abstract "+20% grasp" mixes matched-horizon (**+10 pp**) and longer-horizon (**+20 pp**) — clarify.
3. ViT-g slightly under VJ2 on some globals — say so loudly that ViT-G is required.
4. ADE far from DINOv3-7B (**47.9 vs 55.9**); "SOTA dense" overstated outside depth.
5. Cool-down confounds resolution with LR decay.
6. Nav 10× changes latent **and** DDIM steps — system claim.
7. T1 cumulative, not full factorial — Empiricist must isolate on fixed tiny data.
8. Distillation drops deep SS — dense quality of B/L may lag teacher more than global tables show.

Verdict: Strong methods + transfer paper that closes the dense gap in the JEPA line and re-runs planning with better state. Accept for Accompanying Models as **required** after I-JEPA/V-JEPA 2; when citing robots, demand **T6 matched vs longer-horizon** and **T7 system caveat**.

Reviewer exam bite: Demand **T1+T2+Fig3** before slogan; cite grasp as **60/70/80 (n=10)** not bare "+20%"; punish BEHAVIOR-hybrid-as-result and temporal-mask-as-paper-ablation.

6.4 Empiricist — produced despite empty seat (course smoke; all seats consume)

Seat charge: Course smokes that stress `L_ctx` · λ weighting · deep SS · frozen probes, ranked by GPU honesty. Prefer tiny short clips + ViT-Tiny/S — **not** VisionMix / ViT-G / Franka CEM.

Claim under measurement

Matched short pretrain: weighted `L_ctx` improves a dense proxy vs masked-only; without deep SS, global clip accuracy **drops**; multi-level SS **recovers** globals while keeping most dense gains (directional T1/T2) — without claiming paper absolute %.

Hypotheses (falsifiable — from DELIVERABLE)

1. **H-dense:** Weighted `L_ctx` improves dense proxy (toy depth / patch-neighbor / mini-seg) vs masked-only.
2. **H-tradeoff:** Without deep SS, `L_ctx` **hurts** global clip classification (SSv2-lite / Kinetics-tiny).

3. **H-deep**: Multi-level supervision recovers global accuracy while keeping most dense gains.
4. **H-temporal (extension only)**: Stronger temporal tube masking *may* matter for motion probes — **not a paper ablation**.
5. **H-prior**: Frozen 2.1 features beat V-JEPA 2 on dense probes when public ckpts exist (T8 sign).
6. **H-transfer (E8)**: Better dense features → lower latent goal-distance / higher proxy grasp on **sim** Push-T / LIBERO-lite / DROID stills — **not paper** Franka 80%.

Falsifiers: L_ctx >/masked-only on dense; no global drop without deep SS; deep SS fails to recover; public 2.1 >2 on dense with no caveat; smoke claimed as Franka 80%.

Prioritized ablation matrix (Ablation Bot + DELIVERABLE)

#	Experiment	Paper hook	Smoke	Pri
E1	L_ctx on vs masked-only	T1, Fig 3	Y — PRIMARY	P0
E2	λ fixed / warmup / weighted lite	T2	Partial	P0
E3	Deep SS levels {1,4}	T1, T13	Y — PRIMARY	P0
E4	Multi-modal tokenizer	T1	Partial	P1
E5	Temporal mask (extension)	—	Partial	P2
E6/E7	Frozen 2 vs 2.1 probes	T4/T5/T8/T9	Y / Partial	P0
E8	Tiny AC / embedding nav proxy	T6/T7	Partial	P1
—	VisionMix / cool-down / ViT-G / Franka / NWM	T1/T6/T7/T14	N	Skip

Min package: E1 + E3 + E7, then E2 if GPU free; E8 as joint measurement with public weights. Label any E5 extension.

Exact reproduce stack (EDIT)

```
# Official
git clone https://github.com/facebookresearch/vjepa2
# Pin commit → logs/pins.txt

# Course scout env (quiet outlines – preferred)
export VJEPa21_WORK_ROOT=$SCRATCH/vjepa21-smoke
cd /workspace/vjepa21-quiet
bash env_outline/setup_mamba.sh
# mamba activate vjepa21-smoke
mkdir -p logs runs checkpoints configs
# EDIT partition/account in slurm_*.sh
# Outlines only – do NOT sbatch from quiet box
# NEVER claim paper NYU 0.307 / STA 7.71 / Franka 80% / 10× nav from smoke
```

Run cards (from DELIVERABLE)

E1 — lctx_on_off (PRIMARY)

Field	Spec
Paper	ADE 22.2 → 33.8; NYU 0.682 → 0.474; SSv2 72.8 → 62.5
Goal	Dense proxy ↑; global may ↓ without deep SS
Train	ViT-S/Tiny; 4–8f@128–224; BS 8–16; ~5k–30k steps
Pass	Directional T1; not absolute ADE/NYU
Budget	~12–36 GPU-h · VRAM 12–22 GB
Script	env_outline/slurm_e1_lctx.sh

E3 — deepss_levels (PRIMARY)

Field	Spec
Paper	SSv2 62.5 → 72.1; ADE 33.8 → 38.6
Goal	Global recovers when levels↑; dense mostly kept
Pass	Levels {1,4} logged; directional T1/T13
Budget	+0.5–1× E1
Script	env_outline/slurm_e3_deepss.sh

E7 — frozen_probe (PRIMARY readout)

Field	Spec
Paper	T4/T5/T8/T9 protocol
Goal	Public 2.1 vs 2 same-sign dense gap on mini probes
Pass	Probe curve logged; disclaimer vs paper ViT-G
Budget	1–6 GPU-h (frozen)
Script	env_outline/slurm_e7_frozen_probe.sh

E2 / E5 / E8 — follow-ons

E2 λ mini-grid (Partial); E5 temporal-mask **extension only**; E8 sim AC / embedding-distance proxy — ties Accompanying Models → robotics **without** Franka queue.

Numbers to memorize (exam)

Fact	Value
ADE recipe T1	22.2 / 33.8 / 38.6 / 47.9
NYU recipe T1	0.682 / 0.474 / 0.463 / 0.307
SSv2 recipe T1	72.8 / 62.5 / 72.1 / 77.7
T2 weighted+warmup	ADE 33.8 / SSv2 62.5
Ego4D STA ViT-G	7.71
EK Action ViT-G	40.8
Grasp T6	60 / 70 / 80 (n=10)
Nav T7	ATE 5.687; 10.6 s vs 103.2 s
NYU vs DINOv3-7B	0.307 vs 0.309
Distill ViT-L SSv2 / ADE	76.5 / 46.7

Failure modes cheat-sheet

Symptom	Likely cause	Next action
No dense lift from L_ctx	Steps too few / clips too tiny	Extend steps; try denser stills mix
Global doesn't drop	λ too small / data too easy	Raise λ; check mask coverage
Deep SS no recovery	Levels miswired / fused wrong	Unit-test multi-level shapes
OOM	Paper BS / long clips	Cut frames+res first; BS 8
Claimed 0.307 / 80% from smoke	Overclaim	Report directional only
Temporal-mask logged as paper abl.	Mislabel	Relabel extension

What not to invent / claim

- **Do not** claim course smoke = NYU 0.307 / STA 7.71 / Franka 80% / 10× nav.

- **Do not** claim BEHAVIOR q-score ↑ from citing dense numbers.
- **Do not** launch VisionMix / ViT-G / Franka CEM from course queue.
- **Do not** claim temporal-mask as paper ablation.
- **Do not** claim π_0 /OpenVLA use V-JEPA towers.

Empiricist exam bite: "Mur-Labadia'26: L_ctx ADE 22.2 → 33.8 / SSv2 -10; deep SS restores 72.1; Empiricist E1+E3+E7 on tiny clips for **direction** — never claim ViT-G / Franka / 10× from smoke."

6.5 Visionary — produced despite empty seat (BEHAVIOR recipe; no Discord/Drive)

BEHAVIOR connection:
2025 BEHAVIOR winners ride $\pi_{0.5}$ + **flow-matching action experts** + **SigLIP/PaliGemma** — generative short-horizon motor interface. V-JEPA 2 / 2.1 / 2-AC ≈ **latent predictive imagination** + CEM / diffusion planning with image goals and almost no task demos. **Dense unlock matters:** household tasks need contact geometry, object boundaries, depth for grasp affordances — exactly what V-JEPA 2 lacked and 2.1 adds (NYU 0.307, VOC 85.0, tracking ~70 J&F).
Proposed hybrid (do NOT invent as existing / in-paper result): V-JEPA 2.1 latent rollout for long-horizon feasibility / goal imagination + FM action expert for motor execution. Research agenda only.

BEHAVIOR need	V-JEPA 2.1-shaped response
Visual trunk choice	Default stays SigLIP-class ; 2.1 = video/dense/WM ablation
Depth / contact geometry	Frozen 2.1 features where image SSL lacks temporal consistency
Long-horizon imagination	AC / CDiT latent planners on 2.1 reps (cite T6/T7 carefully)
Goal-image planning	CEM on action-cond. latents (2-AC recipe + denser encoder)
Language instructions	Keep SigLIP parallel / projector — JEPA unlabeled
Eval honesty	Encoder+planner ablations on challenge data — citing 0.307 ≠ q-score

CLEAR: Dense+temporal JEPA is a viable SSL path that upgrades Meta's planning demos.
TENTATIVE: Official BEHAVIOR'26 baseline ships JEPA trunk; hybrid JEPA-rollout + FM wins organizer board.

2026 BEHAVIOR video backbone / world-model recipe (7 bets)

1. **Default language-aligned trunk stays SigLIP-class** (π_{05} -quiet); V-JEPA 2.1 is the **video/dense/WM ablation**, not the silent default.
2. **When to try 2.1:** depth-sensitive manip, contact-rich stages, short-horizon visual-goal MPC, multi-frame tracking — validate with Empiricist frozen probes first.
3. **Prefer frozen ViT-L/B distill** → **LoRA** over course-scale video pretrain; VisionMix is politics.
4. **Pair modalities:** JEPA lacks native language — add projector / keep SigLIP parallel.
5. **World-model path is this paper's strength:** AC + latent nav are "stage imagination" candidates; still need stage structure / RLC-style wrappers from elsewhere.
6. **Honesty:** citing Ego4D 7.71 or Franka +20 pp does **not** move BEHAVIOR q-score.
7. **Smoke ladder:** I-JEPA mask lessons (i_{jepa} -quiet) → 2.1 L_ctx +deepSS tiny clips → frozen household probes → tiny AC head — never jump to ViT-G.

Follow-up research

1. Action-conditioned JEPA on OXE / DROID / BEHAVIOR-1K with 2.1 encoder — sample efficiency vs π_0 BC.
2. Goal-image CEM inside BEHAVIOR scenes vs language-goal VLAs.
3. Dense aux heads (depth/seg/track) for WM planning.
4. Cross-embodied latents (camera + proprio).
5. Uncertainty-aware predictors for contact-rich failure anticipation.
6. Scale / denser WM heads (paper §5 future work).

New applications (beyond Discord/Drive)

- Zero-shot rearrange with **image goals** on new kitchens (2-AC + denser encoder).
- Fast latent navigation for indoor robots (~10× vs VAE latents).
- Video anticipation / safety monitors (predictor residual as surprise; STA-style).
- Simulation-light world models for household robotics pretrain.
- Educational: “representations should be predictive **and** locally grounded.”

Visionary one-liner: V-JEPA 2.1 is the **dense-state upgrade** to Meta’s video world-model line — the piece that makes latent planning spatially trustworthy for grasping and fast navigation; for BEHAVIOR’26 treat it as imagination / dense backbone **ablation** against SigLIP/DINOv3/V-JEPA 2 while policy glue stays $\pi_{0.5}$ / N1.7 — **hybrid is proposal, not result.**

Visionary exam bite: "For BEHAVIOR, default **SigLIP**; ablate **V-JEPA 2.1** dense/WM features via Empiricist probes; cite T6/T7 carefully (n=10; system 10×); **do not** invent JEPA+FM hybrid as in-paper — **no Discord/Drive.**"

(No Discord / Drive checklist content.)

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export VJEP21_WORK_ROOT=$SCRATCH/vjepa21-smoke
cd /workspace/vjepa21-quiet
bash env_outline/setup_mamba.sh
# mamba activate vjepa21-smoke
mkdir -p logs runs checkpoints configs
# EDIT: partition/account in slurm_*.sh
# Pin facebookresearch/vjepa2 commit -> logs/pins.txt
# Outlines only - do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	vjepa21-smoke sketch
env_outline/conda_env.md	Human-readable mamba steps
env_outline/setup_mamba.sh	Create env + clone pointer
env_outline/run_checklist.md	GPU honesty preflight
env_outline/slurm_e1_lctx.sh	Context-loss on/off — E1 PRIMARY
env_outline/slurm_e3_deepss.sh	Deep SS levels — E3 PRIMARY
env_outline/slurm_e5_temporal_mask.sh	Temporal mask extension
env_outline/slurm_e7_frozen_probe.sh	Frozen probes — E7 PRIMARY
env_outline/README.md	Index + honesty rules; siblings

GPU / RAM / TIME budget honesty

Assumptions: ViT-Tiny/S, 4–8f@128–224, BS 8–16 — **not** ViT-G / VisionMix / 64f cool-down / Franka.

Workload	GPU VRAM (peak)	Host RAM	Wall time (order)	Smoke
E1 L_ctx on/off	12–22 GB	64 GB	12–36 h	Y
E3 deep-SS ×{1,4}	same→+20%	64 GB	+0.5–1× E1	Y
E2 λ mini-grid	same	64 GB	overnight	Partial
E5 temporal mask ×3 (extension)	same	64 GB	+few h each	Partial
E7 frozen probe (public ckpt)	6–16 GB	32–64 GB	1–6 h	Y
E4 / E8 tokenizer / AC proxy	8–24 GB	64 GB	overnight–day	Partial
Paper ViT-L 135k VisionMix	multi-GPU	node	days	N
Paper ViT-G + cool-down	multi-node	—	week-class	N
Franka CEM eval	lab + 1×A100	—	lab schedule	N

Honesty rule: log (arch,lctx,lambda,deep_levels,frames,res,steps,BS,probe,wall_s,mem) . Cut frames and resolution before lying about budgets — video ≫ I-JEPA image smoke. Never amortize paper 0.307 / 7.71 / 80% / 10× onto a Partial stub.

Course envelope: target ≤ ~60–90 GPU-h for E1+E3+E7 (+E2); add E8 if public weights cached. Paper ViT-G / Franka story = cite only.

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
L_ctx unlocks dense (T1/Fig3)	E1: dense proxy ↑ vs masked-only	Matching ADE 33.8 / NYU 0.474
Dense hurts global until deep SS	E1→E3: global recovers	Matching 72.8 / 77.7 SSv2
Weighted L_ctx best trade-off (T2)	E2 direction: weighted ≥ fixed mid	Full λ schedule fidelity
Beats V-JEPA 2 on dense (T8)	E7: same-sign gap on mini probes	0.307 RMSE / 47.9 mIoU
+20 pp grasp / better planning (T6)	E8: proxy ↑ or document blocker	Franka 80% / CEM wall-clock
10× faster nav (T7)	Cite; optional embedding CEM toy	5.687 ATE / 10.6 s
SOTA Ego4D / EK	Cite frozen-ckpt if available	Full STA/EK training
Scales to ViT-G + cool-down	Cite only	Re-running it

Pass: Empiricist executes E1 + E3 + E7 with paper-aligned **directional** rankings + GPU honesty logged.
Fail / overclaim: "reproduced Mur-Labadia’26 / matched 0.307 / verified Franka 80% / 10× nav / BEHAVIOR win" from smoke alone.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Archaeologist	2 · LONGEST · ~16	CLEAR/TENTATIVE lineage; genealogy; consistency triangle; inventum; T6/T7 fence; exam bite
Stakeholder	empty seat · ~12–14	Buy dense+WM track; risks (scale, n=10, ADE gap); decision ask
Reviewer	empty seat · ~12–14	T1+T2+Fig3; grasp 60/70/80 clarity; verdict Accept
Empiricist	empty seat · ~12–14	H-dense/H-tradeoff/H-deep; E1+E3+E7; GPU honesty; exam numbers
Visionary	empty seat · ~12–14	BEHAVIOR SigLIP default + 2.1 ablation; hybrid = proposal; no Discord/Drive

Spine (paste into Slide Maker) — 5 bullets:

1. **Unlock dense video features via L_ctx + deep SS** — supervise visible tokens (weighted) + multi-level prediction; keep global motion (SSv2 77.7).
2. **Must-cite:** NYU 0.307; ADE 22.2 → 33.8 → 38.6 → 47.9; Ego4D STA 7.71; grasp 60 / 70 / 80 (n=10); nav ATE 5.687 + 10.6 s vs 103.2 s; L_ctx vs deep SS trade-off (SSv2 72.8 → 62.5 → 72.1).
3. **SSL / WM triangle:** DINOv3/SigLIP (JEA towers) \j MAE/VideoMAE (pixel MIM) \j I-JEPA → V-JEPA → 2 → 2.1 → 2-AC (predictive emb / dense WM).

4. **Lineage (Archaeologist=2 Lukas LONGEST):** CLEAR I-JEPA → V-JEPA → V-JEPA 2 / 2-AC → **2.1** → 2-AC on 2.1 encoder — robotics inventum = **state estimation**, not new planner; cite T6/T7 carefully.
5. **BEHAVIOR’26 + honesty:** default SigLIP; 2.1 = dense/WM ablation; Empiricist E1+E3+E7 directional only; **do not invent JEPA+FM hybrid as in-paper — no Discord/Drive.**

GitHub title sketch: [ECE 605] - V-JEPA 2.1 Archaeologist Making Two

Full Slide Maker brief: /workspace/vjepa21-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/vjepa21-quiet/KANTA_PACK.md
PDF	/workspace/vjepa21-quiet/KANTA_PACK.pdf
Slide brief	/workspace/vjepa21-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/vjepa21-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/vjepa-2.1-2603.14482-summary.md
Ablations	/workspace/papers/vjepa-2.1-2603.14482-ablations.md (AUTHORITATIVE)
Paper notes	/workspace/vjepa21-quiet/paper_notes.md
Env outlines	/workspace/vjepa21-quiet/env_outline/
Paper PDF	/workspace/papers/vjepa-2.1-2603.14482.pdf
Page figs	/workspace/papers/vjepa21-figs/page-*.png
Lineage sibling	/workspace/ijepa-quiet/

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 — Nov 30 Accompanying Models · Archaeologist=2 (Lukas) LONGEST · Kanta NOT seated · all five produced · Ablation Bot AUTHORITATIVE · Tables 6–7 n=10 careful · no Discord/Drive · no invented BEHAVIOR hybrid as in-paper.